

Basics

1. WAP to swap two numbers without using third variable
2. WAP to find maximum of two numbers without using if-else
3. WAP to check even/odd without using % operator
4. WAP to calculate power (a^b) without using built-in functions
5. WAP to reverse a number without using loop

Operator restriction

6. Multiply two numbers without using *
7. Divide two numbers without using / and %
8. Add two numbers without using + (bitwise use karke)
9. Subtract without using -
10. Find square of number without using * or pow()

Bitwise & logic based

11. Find XOR of two numbers without using XOR operator
12. Check if a number is power of 2 (efficient way)
13. Count number of 1s in binary representation
14. Find the only non-repeating element in array
15. Swap odd and even bits of a number

Pattern + thinking questions

16. Print pattern without using nested loops (challenge)
17. Find missing number in array (1 to n) without sorting
18. Check palindrome number without converting to string
19. Find second largest number without sorting
20. Move all zeros to end without extra array

Real thinking / interview type

21. Find GCD without using built-in function
22. Find LCM without multiplication directly
23. Check prime number efficiently (not brute force)
24. Find factorial without using recursion
25. Convert decimal to binary without using built-in functions

Fun challenges

26. Print numbers from 1 to 100 without using loop
27. Generate Fibonacci without using recursion
28. Find sum of digits without using loop
29. Check Armstrong number without power function
30. Find largest digit in a number without array/string